

Pre-Project Report

Project Number	309
Project Name (Hebrew)	תכנון ובניית מאיץ חומרה למודל טרנספורמר על גבי FPGA
Project Name (English)	Transformer Accelerator on FPGA
Team Members' Names	1. Eliran Turgeman 316372002 2. Shay Rask 314951658
Submission Date	27/11/25
Main Chain	Computer Engineering - Hardware and Chip Design Track
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Project Goal (Objective)

Transformer models, which represent the current state-of-the-art in both Natural Language Processing (NLP) and Computer Vision (through architectures such as Vision Transformers – ViT), are known for their extremely high computational complexity and substantial memory requirements. These intensive demands create significant performance bottlenecks, particularly in scenarios where real-time inference, low latency, and energy-efficient computation are essential. The challenge becomes even more pronounced when targeting edge devices, where available computational resources, power budgets, and memory capacity are considerably limited.

The primary objective of this project is to design, implement, and quantitatively evaluate a complete Hardware Accelerator dedicated to a selected Transformer model, deployed on an FPGA platform. The project aims to provide a practical demonstration of how specialized hardware can overcome the computational challenges inherent to Transformer architectures. By developing a dedicated accelerator, the expectation is to achieve clear and measurable improvements in key performance metrics—specifically Latency, Throughput, and overall Energy Efficiency—when compared to traditional CPU or GPU-based solutions. Ultimately, the goal is to show that a hardware-optimized implementation can significantly enhance inference performance while reducing power consumption, thereby enabling efficient deployment of Transformer-based models in real-world, resource-constrained environments.

Project Deliverables (Scope of Project)

1. **Main Output: Functional Hardware Prototype**

A complete, fully operational hardware prototype implemented as a Bitstream file on an FPGA platform (such as Xilinx or Intel). This Bitstream will contain the full accelerator architecture and will be capable of executing inference operations for a selected Transformer component or model—such as a single Attention mechanism or a compact end-to-end Transformer network. The focus is on achieving correct functionality, stable performance, and hardware behavior that aligns with the design objectives.

2. **Documentation: Comprehensive Final Project Report (Book)**

A detailed and structured final report summarizing the full workflow of the project. This includes background research, design considerations, architectural decisions, implementation stages, testing procedures, and performance evaluation. The document will serve as a complete reference to the project's development process and final results.

3. **Presentations: Midterm and Final Project Presentations**

Two structured presentations—one midterm and one final—designed to communicate the project's progression, main challenges, architectural decisions, performance findings, and final conclusions. These presentations are intended to clearly convey the technical contributions and provide an accessible overview of the hardware accelerator and its capabilities.

4. **Auxiliary Documentation: Full Hardware Micro-Architecture Documentation**

Supplementary technical documentation describing the internal structure and building blocks of the accelerator. This includes detailed block diagrams, descriptions of Processing Elements (PEs), explanations of memory hierarchy and dataflow methods (e.g., Memory Tiling, Systolic Array-based processing), and documentation of the chosen development methodology—whether through High-Level Synthesis (HLS) or lower-level Verilog/VHDL design. This material ensures reproducibility and provides full transparency into the hardware architecture.

Work Plan: Defining Intermediate Milestones

Submission Milestone (Midterm / Final Presentation)	Intermediate Goal (Work Phase)	Brief Task Description	Estimated End Date	Responsible Party
Midterm Presentation	1. Literature Review & Theoretical Foundation	Deep literature review on Transformer architectures, Quantization models for hardware implementation, and advanced FPGA acceleration techniques.	12/25	Eliran Turgeman
Midterm Presentation	2. Architecture Selection and Methodology	Selecting a specific Transformer model defining the quantization strategy and designing the hardware accelerator architecture at the block level.	01/26	Shay Rask
Midterm Presentation	3. Critical Component Implementation / POC	Implementing one critical accelerator component using HLS or Verilog, and initial functional verification (Proof of Concept).	02/26	Eliran Turgeman
Preparation	Midterm Presentation Preparation		02/26	Both Students
Final Presentation	4. Full Implementation and System Integration	Completing the development of all processing units and integrating all blocks into a complete accelerator system ready for deployment on the FPGA.	04/26	Shay Rask
Final Presentation	5. Performance Testing and Calibration	Implementing the system on the FPGA, conducting full inference tests, measuring performance metrics (Latency, Throughput), and calibrating the system for optimal hardware resource utilization.	05/26	Eliran Turgeman
Final Presentation	6. Results Analysis and Final Report Writing	Comparing the measured results against CPU/GPU performance, analyzing the advantages and limitations of the implementation, and writing the comprehensive Final Report.	06/26	Both Students
Presentation	Final Presentation & Final Report Submission		06/26	Both Students



Work Plan: Defining Schedule

List of Sources

1. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). **"Attention Is All You Need."** *Advances in Neural Information Processing Systems*.
2. Zhu, D., Li, Y., Chen, J., & Ma, X. (2023). **"High-Performance FPGA Acceleration for Transformer-Based Models: A Survey."** *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*.
3. Li, Q., Chen, S., Cheng, H., & Huang, T. (2022). **"ME-ViT: A Single-Load Memory-Efficient FPGA Accelerator for Vision Transformers."** *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*.
4. Liu, G., & Sun, Y. (2021). **"FTRANS: Energy-Efficient Acceleration of Transformers using FPGA."** *Proceedings of the 2021 ACM/SIGDA International Symposium on FPGAs (FPGA)*.
5. Cai, C., Zhang, Y., & Chen, Y. (2020). **"A High-Throughput and Energy-Efficient FPGA-Based Transformer Accelerator."** *2020 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*.

Signatures

We hereby confirm that we have read, understood, and commit to meeting the goals, deliverables, and timeline specified in this Pre-Project Report.

Student 1

Student 2

Supervisor

Academic Authority

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